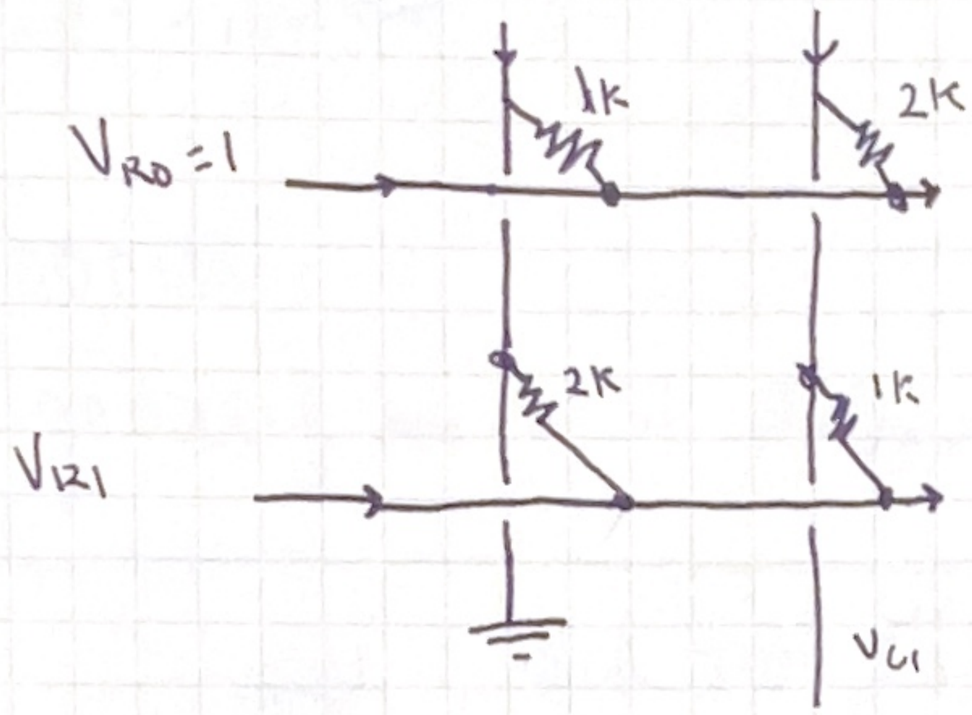
Navid Hamszadeh

ECE 510 S26 CMAN 6

1)




$I_{C0} = \frac{1}{1k} \cdot 1V = 1mA$

2) KCL: $V_{C1} - 1 + 2V_{C1} - 2V_{R1} = 0$

$-V_{R1} + 2V_{C1} - 2V_{R1} = 0$

$\Rightarrow 2V_{C1} - 3V_{R1} = 0 \Rightarrow V_{C1} = \frac{3}{2} V_{R1}$

$\Rightarrow 2V_{R1} - \frac{9}{2} V_{R1} = -1 \Rightarrow -\frac{5}{2} V_{R1} = -1 \Rightarrow V_{R1} = \frac{2}{5} = 0.4V$

$\frac{3}{2} \cdot \frac{2}{5} = \frac{6}{10} = V_{C1} = 0.6V$

$I_{C0} = \frac{1-0}{1k} + \frac{0.4-0}{2k} = 1.2mA$

3) Sneak path currents are indistinguishable from signal current at output, and sneak path current grows with respect to array size, i.e. MVM error increases with respect to array size. The additional current adds to the MVM calculation, corrupting the result.